



SARDAR PATEL UNIVERSITY

US03MABCA01 || Fundamentals of Data Structures

Date : 16-10 -2024

Time : 02.00 p.m. – 04.30 p.m.

Total : 50 Marks

Q-1 Select the appropriate option for the following questions: [10]

- 1) The data structures directly operated by machine instruction are called _____.
A. Structured C. Non-Primitive
B. Primitive D. None
- 2) Index is also known as _____.
A. Subscript C. Non-subscript
B. Base D. Type
- 3) An operation that is used to give the value of an element at a particular position from a top of a stack is known as _____.
A. Push C. Pop
B. Peep D. Change
- 4) In the _____ notation the operators are written after the operands.
A. prefix C. postfix
B. infix D. None of above
- 5) _____ is a set of disjoint trees.
A. Path C. Directed tree
B. Forest D. Height
- 6) The number of sub tree of a node in a tree is called _____ of a node.
A. Degree. C. Level.
B. Height. D. Index.
- 7) Which of the following is NOT the type of Linked list?
A. Circular linked list C. Doubly linked list
B. Hybrid linked list D. Singly linked list
- 8) In which searching technique, elements are eliminated by half in each pass...
A. Linear Search C. Binary Search
B. Both A & B D. None of these
- 9) An array is _____ a data structure.
A. Unordered C. Non-composite
B. Linear D. Heterogeneous
- 10) The line formed for paying fees for college is an example of _____.
A. Stack C. Simple Queue
B. Circular Queue D. Priority Queue

Q-2 Explain in brief classification of Data Structure and list applications of Data Structure. [10]

OR

Q-2 What is array? Discuss 1-D array with an example and explain address calculation of elements of one dimensional array in detail. **[10]**

C.P.T.O.)

Seat No : _____

No. of Printed Pages: 2

Q-3 Write an algorithm for Push, POP, Pèep and Change operations of a stack. [10]

OR

Q-3 Define Queue. List all and explain any three types of Queue with example. [10]

Q-4 Write down an algorithm for inorder and preorder traversal of binary tree. [10]

OR

Q-4 List applications of tree. What is Binary tree? Explain the representation of binary tree. [10]

Q-5 Define Linked List. List and explain all types of linked lists with example in detail. [10]

OR

Q-5 What is sorting? List sorting techniques and explain ANY ONE of them with algorithm. [10]

